Abdelrahman Khalifa

Software Engineer | Rust, C++, Linux

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SUMMARY

Software engineer working in Rust, modern C++, C and Python. Built the Printers page for COSMIC Settings in Google Summer of Code 2026 with The Linux Foundation (7 PRs merged upstream) and sole software engineer at EKSON, owning firmware, desktop app and backend. Codeforces Expert, 4-time ECPC finalist.

TECHNICAL SKILLS

Languages: Rust, C++ (modern), C, Python (asyncio), JavaScript (Node.js), SQL, Java, C#, Bash

Computer science: Data structures and algorithms, operating systems, concurrency and multithreading, async/event-driven design, memory management, networking (TCP/UDP, IPC)

Systems & Linux: Linux (Ubuntu/Debian), FFI, IPC (varlink), CUPS/IPP, DNS-SD/mDNS, kernel modules and device drivers, Zephyr RTOS, FreeRTOS, Yocto, Buildroot

Backend & tools: Node.js, Express, MongoDB, REST APIs, PyQt5; Git/GitHub, GitHub Actions CI, Cargo, CMake, testing, debugging, code review

EXPERIENCE

Software Developer (Google Summer of Code) | **The Linux Foundation / OpenPrinting** | May 2026 – Oct 2026

GSoC 2026: Printers page for COSMIC Settings, in Rust; passed the final evaluation | *github.com/Abd002/GSOC-2026*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI: printers grouped by device, status and supply levels, print queues, live updates, and an Add Printer flow for legacy printers (cosmic-settings #2182, in review).
- Designed it as 5 Rust crates and ran the C FFI backend as a varlink IPC service in the settings daemon, so a native crash cannot take down Settings.
- **7 PRs merged upstream**: ported cups-rs to libcups3 with CUPS 2 and 3 builds and CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer | **EKSON** | Aug 2025 – Present

Sole software engineer for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Built the PyQt5 control app; used asyncio to wait for ESP32 and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Built remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware with esptool; GitHub Actions CI.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | Dec 2023 – Jan 2024

- Built Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing.

OPEN SOURCE PROJECTS

Contributor — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977, merged).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553, merged).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214, merged).

PROJECTS

- **IoT Device Communication System** (C++, Raspberry Pi 4, QEMU, Yocto): object-oriented framework for TCP unicast and UDP multicast between a server and emulated clients. GitHub
- **Operating System Simulator** (Java): CLI simulating scheduling, memory management, system calls and locks. GitHub
- **Inventory Management System** (C#, WinForms, SQL): stock, orders and deliveries app with role-based access control and reports. GitHub
- **Team projects (co-developed):** Sarh Al-Tafawuq, a production web platform (live); Nafaqati, a Flutter Android expense tracker (Google Play).

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- IEEE student branch Treasurer: managed a budget of more than $10k.

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | 2020 – 2025

- Grade: Very Good with Honors. Graduation project: AUTOSAR-based V2X vehicle on STM32; built the firmware update path and a dual-bank bootloader with rollback.